Boolean algebra and switching functions exercises

Use three-variable index $4A + 2B + C$. Unless stated otherwise, SOP cost minimizes term count first, then literal occurrences. A factored general expression is not an SOP. These are original instructional exercises.

1. For $f = \bar{A}BC + A\bar{B}C + ABC$, give the canonical minterm indices and a minimal SOP, proving its minimality. Compare the literal count of the factored expression with that of the SOP.
2. Give canonical DNF and canonical CNF for $A \oplus B$, proving equality row by row. Do the two true indices change if the input order is reversed? Explain why the answer for this function does not extend to every function.
3. Simplify $(A + B)(A + \bar{B})(\bar{A} + B)$ using Boolean laws, naming each law. Find the dual and complement of the entire original expression. Do not guess from the simplified string without justification.
4. A three-variable specification has ON set $\{0, 2, 4\}$, don't-care set $\{1, 5\}$ and OFF set $\{3, 6, 7\}$. Find an SOP minimizing total literal count, prove a lower bound, and identify the chosen completion.
5. A function is true exactly at $000, 001, 010, 011$. List its four minterms and locate them on a K-map with rows A and columns BC. Reduce them to one literal by distributivity. Then use rows AB and columns C to explain wrap-around for $\Sigma m(0, 1, 4, 5)$.
6. Prove that in a finite Boolean algebra, if every atom a satisfies $a \leq x$, then $x = 1$. Allow the one-element algebra $0 = 1$, and handle the case of no atoms.
7. For $h = \Sigma m(1, 2, 3, 4, 5, 6)$, list every prime and its covered indices. Prove that there is no essential prime, and find all prime covers with the fewest terms. Explain why each optimal cover also has the least total literal count.
8. Only two-input NAND gates may be used. The inputs provide A, B, C, with no complements. For $f = A\bar{B} + BC$, specify the intermediate signals of a four-gate circuit, give its complete truth table, and find its maximum gate depth. Which gate can be omitted if $\bar{B}$ is already available? You need not prove that four gates are globally minimal.
9. A finite Boolean algebra B has $|B| = 16$. Count its atoms, the elements above a particular atom, and ordered pairs (x, y) with $x \wedge y = 0$. Explain how the representation isomorphism preserves each counted condition.

Boolean algebra and switching functions exercises (continued)

10. In the Boolean algebra of positive divisors of 30, find the complement of 6, $6 \wedge 10$, $6 \vee 10$, and all Boolean-algebra automorphisms sending 6 to 10. Explain why $12/d$ does not automatically give a complement for every element of the divisor lattice of 12.
11. Prove that a nonzero Boolean algebra cannot have exactly three atoms and only six elements. Without finiteness, explain why an algebra with no atoms need not have only one element. Give a concrete Boolean algebra and verify that a smaller nonzero element can always be found.
12. In any Boolean ring with identity, where every $x^2 = x$, prove that the characteristic is 2 unless the ring is the zero ring, and prove commutativity. Express Boolean XOR $x \oplus y$, join $x \vee y$ and complement $\bar{x}$ using ring addition and multiplication. State when OR differs from ring addition.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study